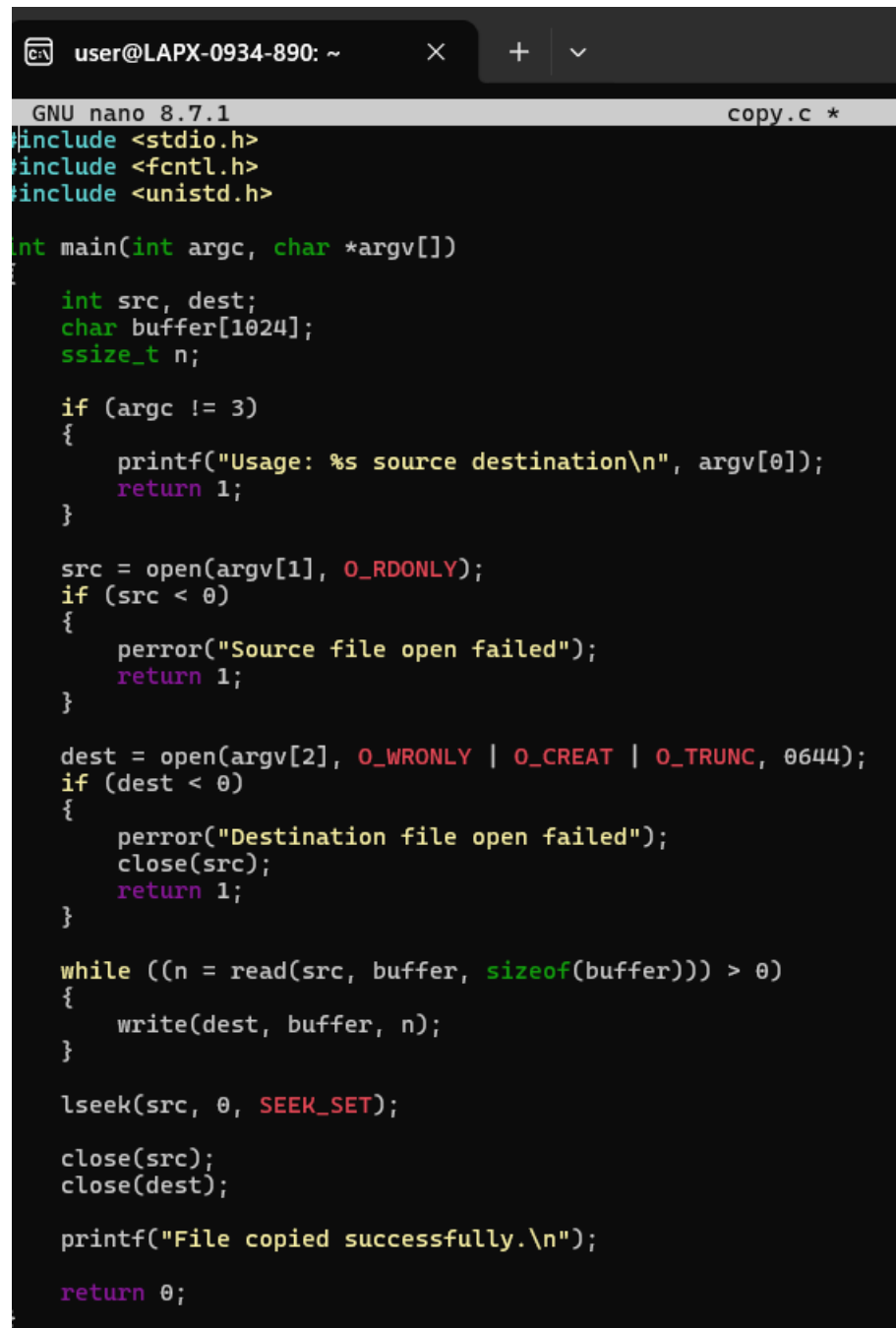


OSSP PRACTICAL WEEK-9

TASK-1: Implement a file copy utility using low-level Linux file I/O system calls such as open(), read(), write(), lseek() and close(). Compare its performance with standard library functions.

SOURCE CODE:



```
GNU nano 8.7.1 copy.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main(int argc, char *argv[])
{
    int src, dest;
    char buffer[1024];
    ssize_t n;

    if (argc != 3)
    {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }

    src = open(argv[1], O_RDONLY);
    if (src < 0)
    {
        perror("Source file open failed");
        return 1;
    }

    dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (dest < 0)
    {
        perror("Destination file open failed");
        close(src);
        return 1;
    }

    while ((n = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, n);
    }

    lseek(src, 0, SEEK_SET);

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

OUTPUT:

```
user@LAPX-0934-890:~$ nano copy.c
user@LAPX-0934-890:~$ gcc copy.c -o copy
user@LAPX-0934-890:~$ echo "Hello Ubuntu Linux File I/O" > input.txt
user@LAPX-0934-890:~$ ./copy input.txt output.txt
File copied successfully.
user@LAPX-0934-890:~$ cat output.txt
Hello Ubuntu Linux File I/O
user@LAPX-0934-890:~$ |
```

OBSERAVTION: The source file was successfully copied to the destination file using low-level Linux file I/O system calls. The contents of both files were verified and found to be the same.

TASK 2 : Implement a file copy utility using low-level Linux file I/O system calls such as open(), read(), write(), lseek() and close(). Compare its performance with standard library functions.

SOURCE CODE :

```
GNU nano 8.7.1 redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int input, output;

    input = open("input.txt", O_RDONLY);
    output = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (input < 0 || output < 0)
    {
        perror("File open failed");
        return 1;
    }

    dup2(input, STDIN_FILENO);
    dup2(output, STDOUT_FILENO);

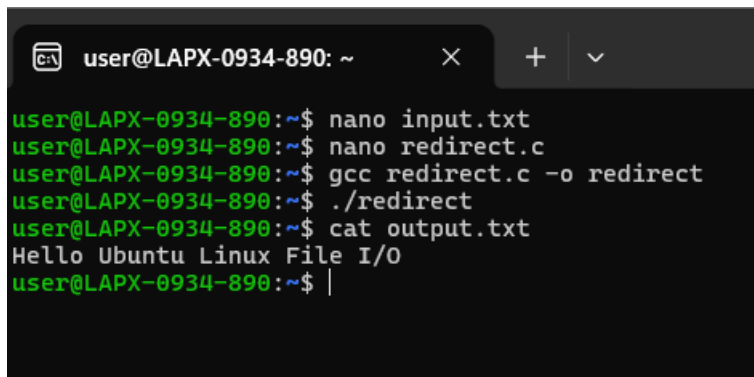
    close(input);
    close(output);

    char buffer[100];

    while (read(STDIN_FILENO, buffer, sizeof(buffer)) > 0)
    {
        write(STDOUT_FILENO, buffer, sizeof(buffer));
    }

    return 0;
}
```

OUTPUT :

A terminal window with a dark background and light-colored text. The window title bar shows 'user@LAPX-0934-890: ~' and standard window controls. The terminal displays a series of commands and their outputs. The commands are: 'nano input.txt', 'nano redirect.c', 'gcc redirect.c -o redirect', './redirect', and 'cat output.txt'. The output of the last command is 'Hello Ubuntu Linux File I/O'.

```
user@LAPX-0934-890: ~  
user@LAPX-0934-890:~$ nano input.txt  
user@LAPX-0934-890:~$ nano redirect.c  
user@LAPX-0934-890:~$ gcc redirect.c -o redirect  
user@LAPX-0934-890:~$ ./redirect  
user@LAPX-0934-890:~$ cat output.txt  
Hello Ubuntu Linux File I/O  
user@LAPX-0934-890:~$ |
```

OBSERVATION : The source file was successfully copied to the destination file using low-level Linux file I/O system calls. The contents of both files were verified and found to be the same.